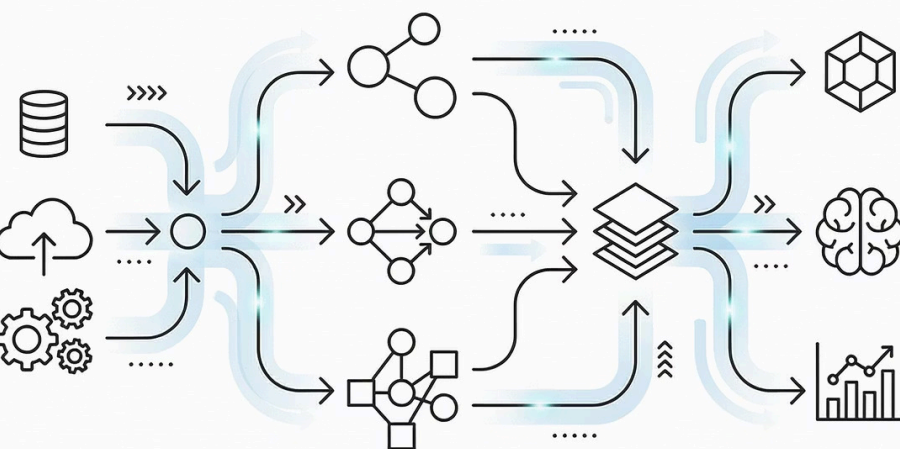
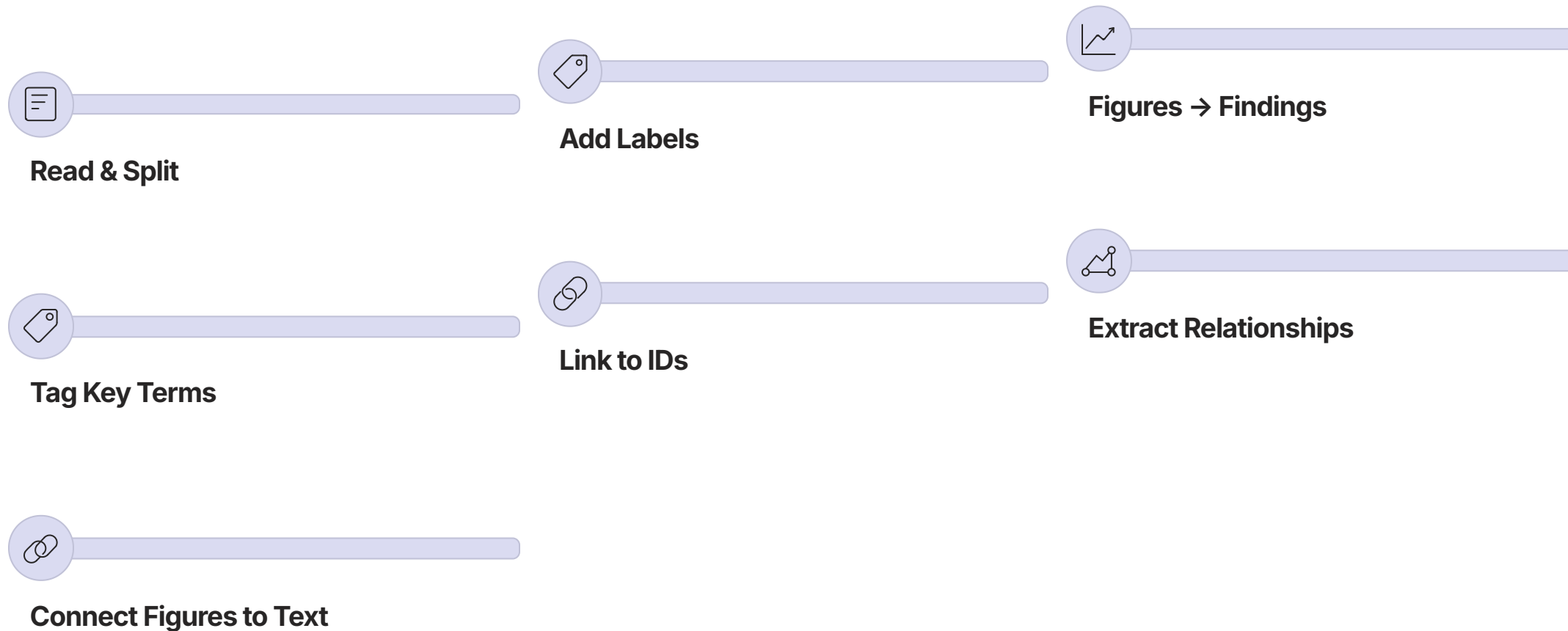




Semantic Content Enrichment

- A seven-step pipeline that transforms publisher XML articles into AI-ready, semantically rich content for intelligent search.

The Seven-Step Pipeline

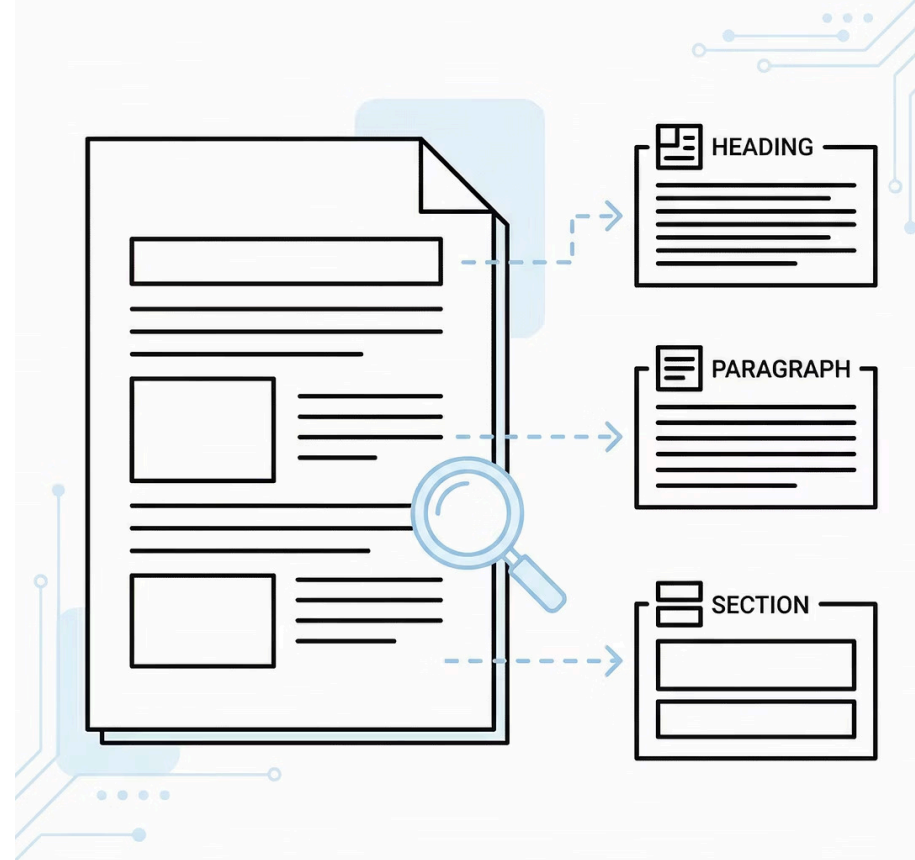


- Each article is parsed, contextualised, and enriched so that every passage is independently searchable.
- Figures and entities are made interoperable for downstream use.

STEP 1

Read & Split

- Master-copied Publisher XML is parsed with a fault-tolerant parser.
- Articles are split into sections and passages.
- Short passages are merged with neighbours.
- A one-sentence overlap carries trailing context forward to prevent cross-boundary evidence loss.



STEP 2

Add Context Labels

- AI writes a short label for every passage stating:

Article

Which paper it is from

Section

Methods vs. Results

Author & Year

First author, publication year

STEP 3

Figures to Findings

- Rewrite each figure and table caption as a plain-language finding with specific values - e.g. "*Treatment X reduced tumour size by 40%*" - replacing blank image placeholders with searchable evidence.

Tag Terms & Link to Standard IDs

What Gets Tagged

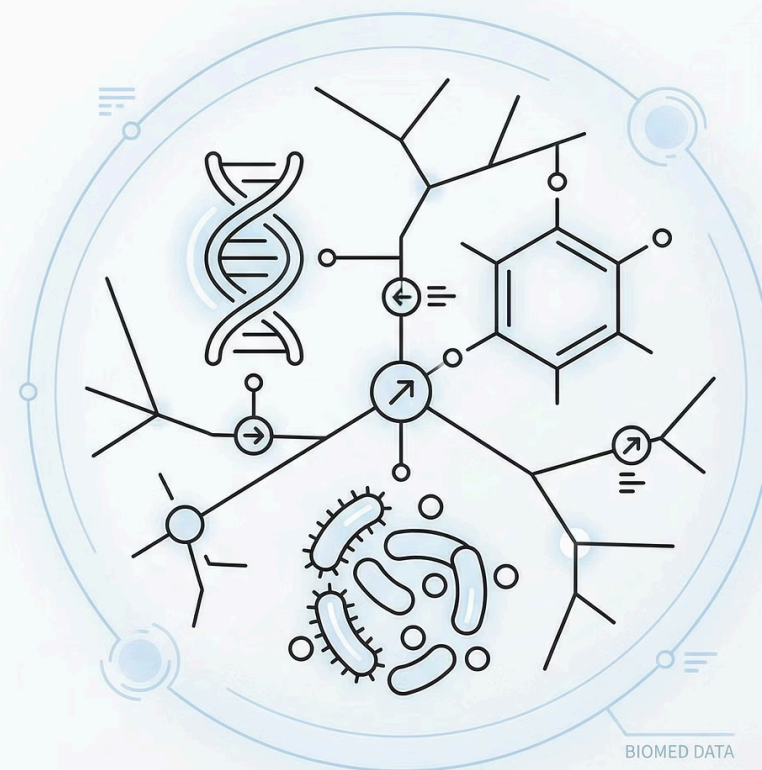
- Genes, diseases, chemicals, cell lines, and organisms in every passage.

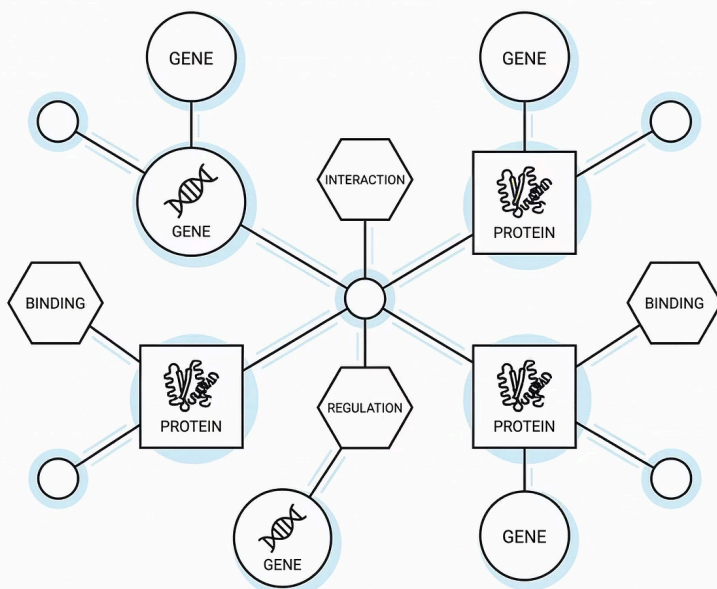
Why IDs Matter

- Each term is matched to a shared ID in NCBI, MeSH, or ChEBI - so *m6A*, *N6-methyladenosine*, and *m⁶A* all resolve to one entity.

Cross-Paper Links

- The same gene across two papers is no longer two unrelated strings - shared IDs connect them corpus-wide.





STEP 6

Extract Relationships

- AI pulls out explicit relationships between terms - e.g. "*Gene X inhibits Protein Y*" - stored as typed triples with an evidence snippet.

No False Co-occurrence

- Two terms in the same passage does not mean they interact - only explicit statements are stored.

Predicate-Aware Search

- "What inhibits X?" matches passages asserting Y inhibits X via a typed predicate field.

STEP 7

Connect Figures to Text

- Passages that mention a figure are linked to it - and the figure links back.
- Evidence locked in a chart is now surfaced when a query matches the surrounding text, and vice versa.



Split Evidence Solved

- Text and figure cross-linked so either surfaces the other.

Orphaned Figures Solved

- Figures with no query match are found via their linked text passage.

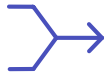
Structured Data Export

- A separate manual step converts the enriched index into **JSON-LD** - each article described using standard web vocabularies, with all entity IDs pointing to live database records.



Interoperable

- Enrichment is no longer locked inside the pipeline.



Deduplicated

- Same entity across 12 passages produces one canonical record.



Live IDs

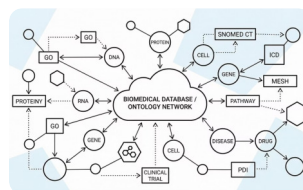
- Canonical URLs resolve to live ontology records.

22 Failure Modes — All Addressed



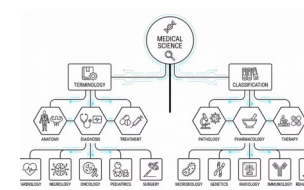
Reading & Splitting

- Custom fault-tolerant parser, passage merging, and one-sentence overlap handle DTD failures, short passages, and cross-boundary evidence loss.



Term Tagging & IDs

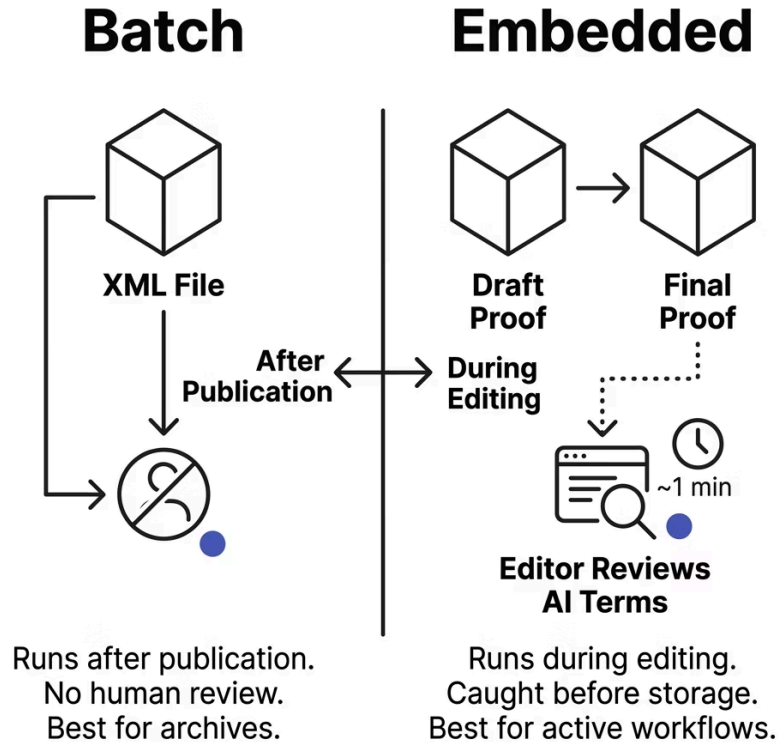
- Spelling variants, opaque abbreviations, and cross-paper duplicates all resolved via shared database IDs.



Vocabulary & Hierarchy

- Medical vocabulary hierarchy matches broad queries to specific article terms — normalising upward to shared parent concepts.

Batch vs. Embedded Workflow



The One Meaningful Difference

- AI occasionally mislabels terms - e.g. an imaging score tagged as a gene.
- The **batch pipeline** has no gate; bad labels go straight into the index.
- The **embedded pipeline** adds a light weight editor review that filters these out before storage.

Of 22 failure modes: **21** handled identically in both · **1** handled better in embedded · **3** (structured export) require a separate manual step in both.